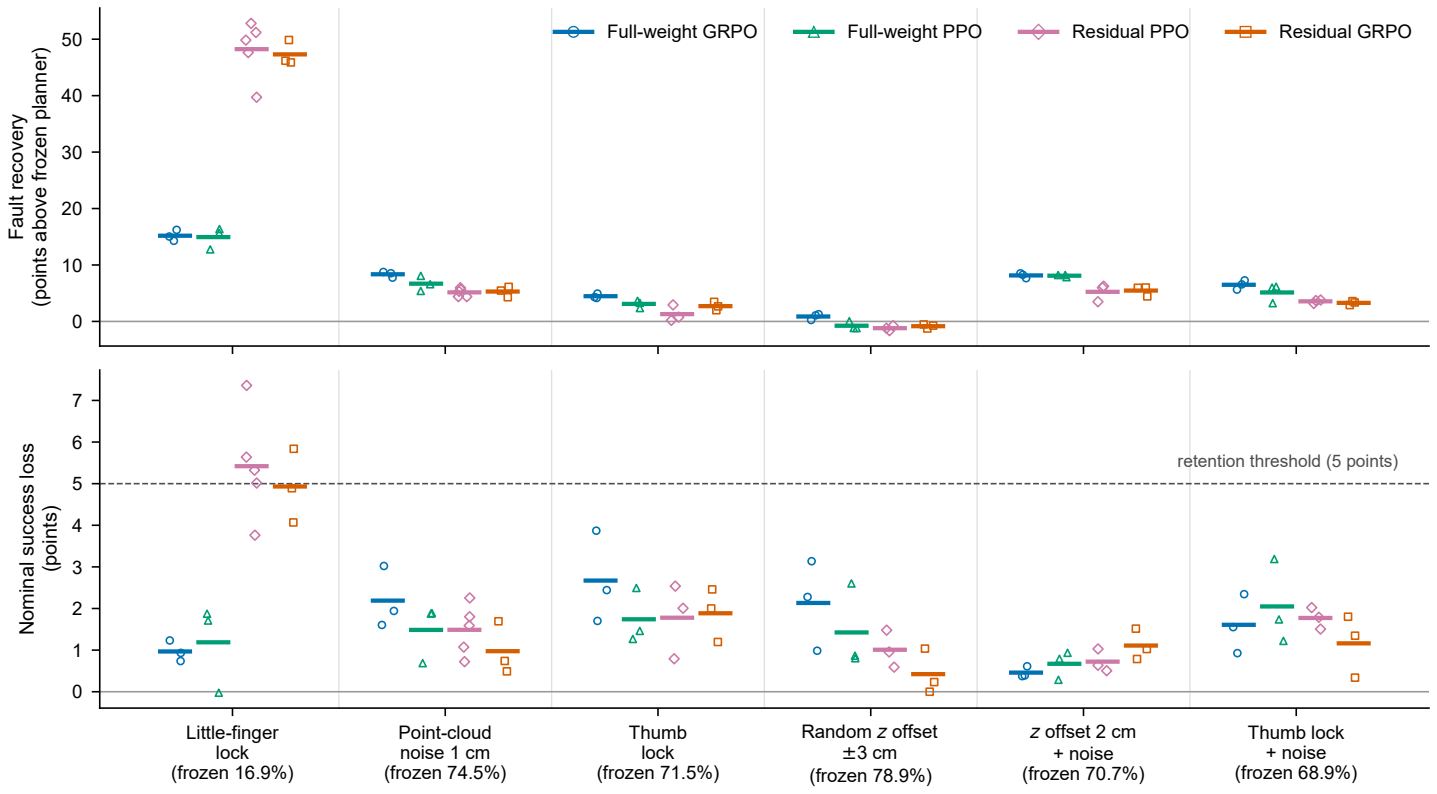
Full-weight GRPO
Full-weight PPO
Residual PPO
Residual GRPO
Fault recovery
(points above frozen planner)
50
40
30
20
10
0
Nominal success loss
(points)
7
6
5
4
3
2
1
0
retention threshold (5 points)
Little-finger
lock
(frozen 16.9%)
Point-cloud
noise 1 cm
(frozen 74.5%)
Thumb
lock
(frozen 71.5%)
Random z offset
±3 cm
(frozen 78.9%)
z offset 2 cm
+ noise
(frozen 70.7%)
Thumb lock
+ noise
(frozen 68.9%)